Oceanic storms

Although open oceans are swept by winds from one direction most of the time, they are also affected by local weather systems that change the wind pattern, and often bring heavy rain. These weather systems are generated by warm, moist air rising off the oceans. They form zones of circulating air called cyclones, which can cause destructive storms.

STORM CLOUDS

Warm air rising from sun-warmed oceans carries a lot of invisible water vapor with it. As it rises, the air cools. This makes some of the vapor turn back into the tiny water droplets that form clouds. Where a lot of warm, moist air is rising, this process builds up giant storm clouds that contain a huge weight of water. Eventually, the water spills out of them as heavy rain.

SWIRLING CYCLONES

As warm air rises, it reduces the weight of air at sea level, creating a zone of low air pressure. Surrounding air swirls into the low-pressure zone to replace the rising air. The faster the warm air rises and the lower the pressure, the faster more air moves in, causing strong winds. These weather systems are called cyclones or depressions. They swirl counterclockwise in the northern hemisphere, and clockwise in the southern hemisphere.